

Análisis Matemático de Algoritmo

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Resumen Ejecutivo

Este documento presenta la traducción completa de un algoritmo computacional a tres formas matemáticas distintas. Las ecuaciones resultantes, debido a la complejidad logarítmica del algoritmo original (Miller-Rabin), son extensas pero polinomialmente acotadas.

Part I

Función de Transición

$$\begin{aligned}n[t + 1] &= n \\result[t + 1] &= 0\end{aligned}$$

Part II

Conversión a Polinomio

Ecuación Unificada P=0

$$\begin{aligned}&(e_0 \cdot (1 - e_0))^2 + (e_0 \cdot ((1) - (x) - (e_1^2 \\&+ e_2^2 + e_3^2 + e_4^2)))^2 \\&+ ((1 - e_0) \cdot ((x) - (1) - 1 - (e_5^2 + e_6^2 + e_7^2 \\&+ e_8^2)))^2 + (e_{11} - (x - 1))^2 \\&+ (P_{fib}(fib, fib, e_{11}, e_{10}))^2 + (e_{13} - (x - 2))^2 \\&+ (P_{fib}(fib, fib, e_{13}, e_{12}))^2 + (e_9 - (e_{10} + e_{12}))^2 \\&+ (RET - ((e_0) \cdot (x) + (1 - e_0) \cdot (e_9)))^2 \\&+ (n[t + 1] - (n))^2 + (result[t + 1] - (0))^2 = 0\end{aligned}$$

Part III

Fórmulas Generadoras

$$\begin{aligned}G &= 1 - ((e_0 \cdot (1 - e_0))^2 + (e_0 \cdot ((1) - (x) - (e_1^2 \\&+ e_2^2 + e_3^2 + e_4^2)))^2\end{aligned}$$

$$\begin{aligned}
& + ((1 - e_0) \cdot ((x) - (1) - 1 - (e_5^2 + e_6^2 + e_7^2 \\
& + e_8^2)))^2 + (e_{11} - (x - 1))^2 \\
& + (P_{fib}(fib, fib, e_{11}, e_{10}))^2 + (e_{13} - (x - 2))^2 \\
& + (P_{fib}(fib, fib, e_{13}, e_{12}))^2 + (e_9 - (e_{10} + e_{12}))^2 \\
& + (RET - ((e_0) \cdot (x) + (1 - e_0) \cdot (e_9)))^2 \\
& + (n[t + 1] - (n))^2 + (result[t + 1] - (0))^2
\end{aligned}$$

Part IV

Fórmulas Matemáticas (Estilo Willans)

1 Representación Simbólica

$$\begin{aligned}
f(N) &= \sum_{R=0}^{\infty} R \cdot \lfloor \frac{1}{1 + \mathcal{D}_{sym}} \rfloor \\
&\mathcal{D}_{sym} \\
&= (\epsilon_0 \\
&\mathit{mathitcdot}(1 - \epsilon_0))^2 + (\epsilon_0 \\
&\mathit{mathitcdot}((1) - (x) - (\epsilon_1^2 + \epsilon_2^2 + \epsilon_3^2 + \epsilon_4^2)))^2 + \\
&((1 - \epsilon_0) \\
&\mathit{mathitcdot}((x) - (1) - 1 - (\epsilon_5^2 + \epsilon_6^2 + \epsilon_7^2 + \epsilon_8^2)))^2 + \\
&(\epsilon_{11} - (x - 1))^2 + (\Phi_{fib}(fib, fib, \epsilon_{11}, \epsilon_{10}))^2 + \\
&(\epsilon_{13} - (x - 2))^2 + (\Phi_{fib}(fib, fib, \epsilon_{13}, \epsilon_{12}))^2 + \\
&(\epsilon_9 - (\epsilon_{10} + \epsilon_{12}))^2 + (RET - ((\epsilon_0) \\
&\mathit{mathitcdot}(x) + (1 - \epsilon_0) \\
&\mathit{mathitcdot}(\epsilon_9)))^2 + \\
&(n - (n))^2 + (R - (0))^2
\end{aligned}$$

2 Representación Operacional

$$\begin{aligned} f(N) &= \sum_{R=0}^{\infty} R \cdot \left[\frac{1}{1 + \mathcal{D}(N, R, \mathbf{x})} \right] \\ &= \sum_{\mathbf{x} \in \mathbb{N}^{20}} \left((x_{133}(1 - x_{13}))^2 + (x_{133}((1) - (x_{19}) - (x_7^2 + x_{10}^2 + x_{15}^2 + x_{18}^2)))^2 \right. \\ &\quad \left. + ((1 - x_{13})_3((x_{19}) - (1) - 1 - (x_{11}^2 + x_{12}^2 + x_{14}^2 + x_9^2)))^2 + (x_6 - (x_{19} - 1))^2 + (x_1(x_8, x_8, x_6, x_5))^2 \right. \\ &\quad \left. + (x_2 - (x_{19} - 2))^2 + (x_1(x_8, x_8, x_2, x_4))^2 + (x_{17} - (x_5 + x_4))^2 \right. \\ &\quad \left. + (x_{16} - ((x_{13})_3(x_{19}) + (1 - x_{13})_3(x_{17})))^2 + (x_{20} - (x_{20}))^2 + (R - (0))^2 \right) \end{aligned}$$

Vector de estado: $\mathbf{x} \in \mathbb{R}^{20}$ (variables internas anonimizadas).

Part V

Sistema Dinámico (Recurrencia Infinita)

Sistema de recurrencia generado (Placeholder)